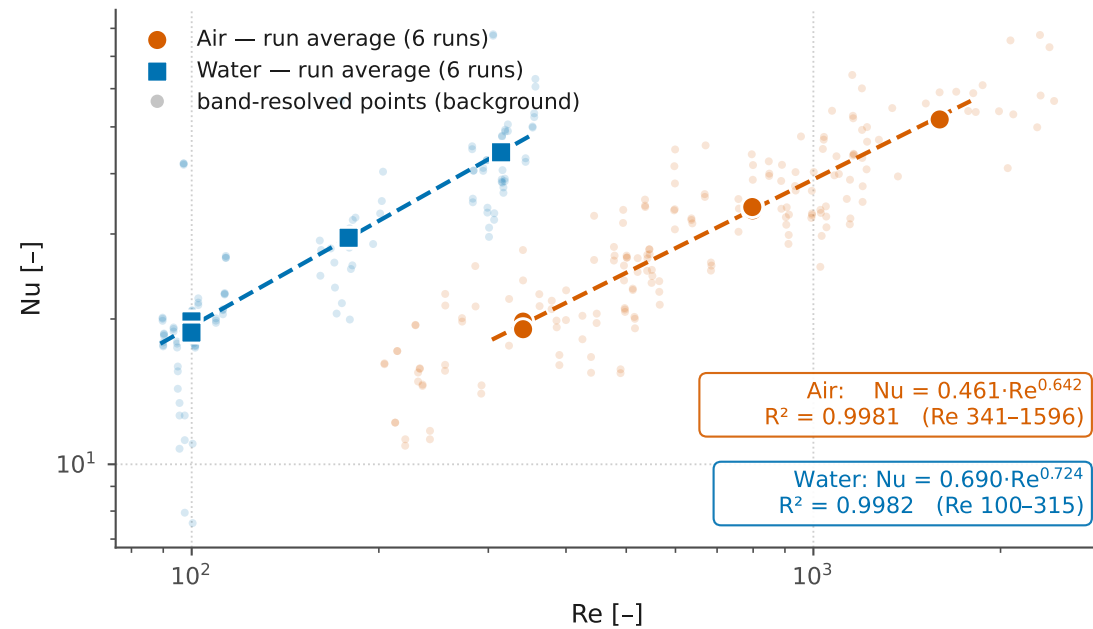
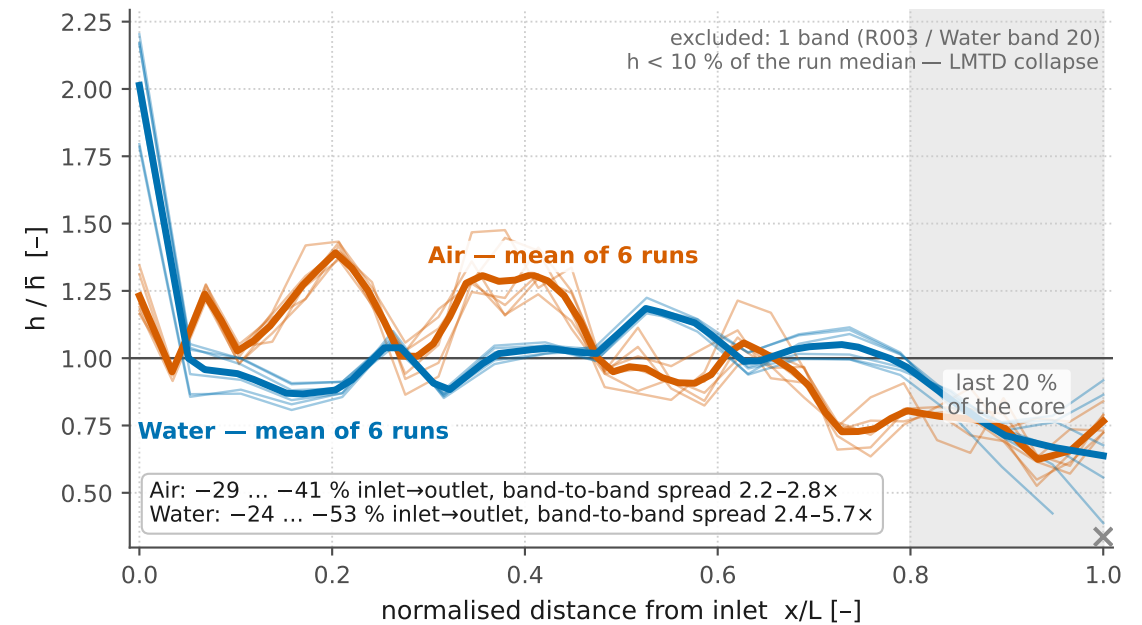


Band-resolved analysis of a TPMS gyroid heat exchanger: from six flow regimes to a design correlation set

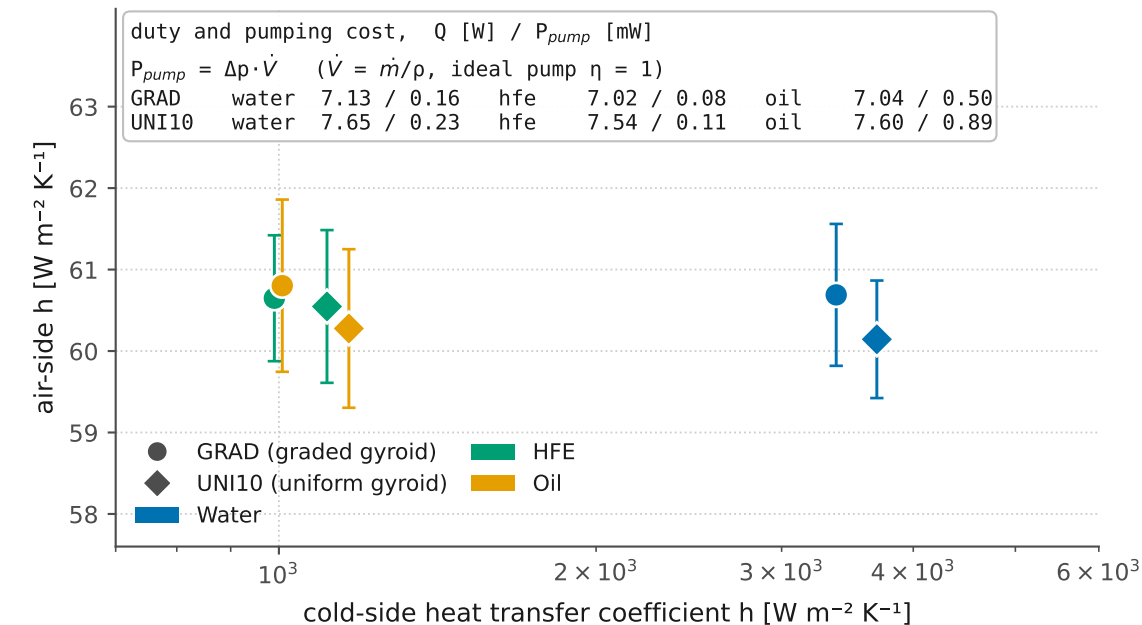
(a) Design correlation $Nu(Re)$ — both streams



(b) The mean hides a 2-6× band-to-band variation

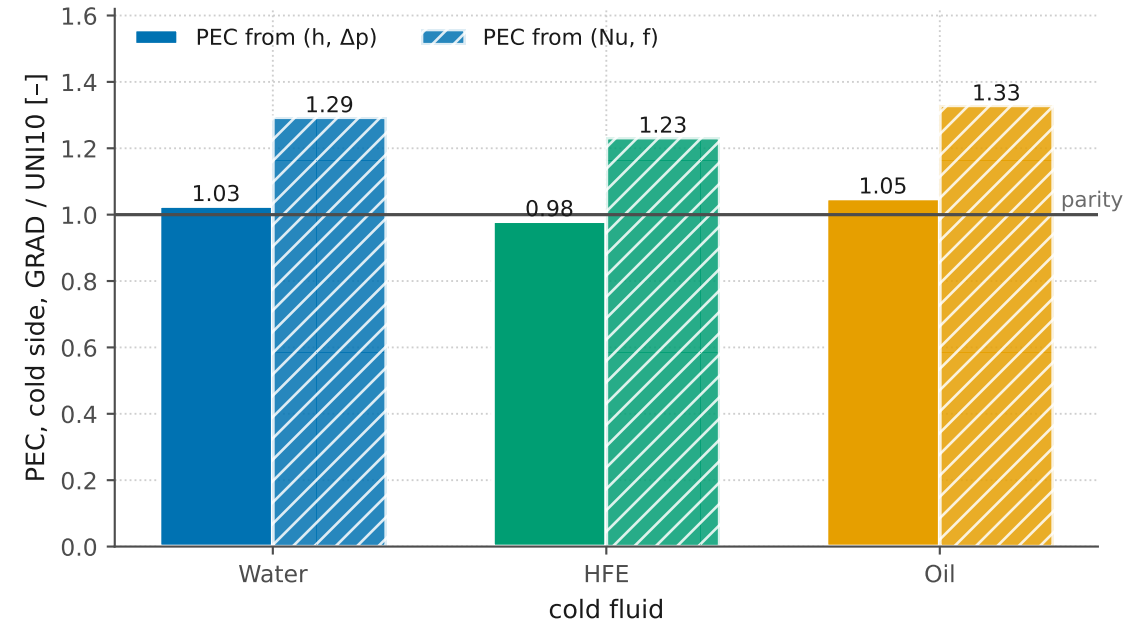


(c) Cold-fluid choice sets the pumping cost, not the duty



cold-side h varies $3.7\times$ → duty stays at $Q = 7.02\text{--}7.65$ W;
on UNI10 oil costs $3.9\times$ the pumping power of water for the same duty

(d) Verdict at equal pumping power — and the validation behind it



$PEC = (X_G/X_U)/(Y_G/Y_U)^{1/3}$, $X, Y = (h, \Delta p)$ or (Nu, f) , $G/U = GRAD/UNI10$; the $1/3$ exponent ⇒ equal pumping power
Validation (transient window 18–19 s, 50 time steps)
energy balance $|Q_{air} - Q_{cold}|$: 0.33–0.67 % mean, ≤ 1.5 % worst step
temporal scatter of h : air 1.2–1.7 %, cold liquid 0.02–1.14 %